ABHISHEK SHRIVASTAVA

Senior Data Engineer → AI/ML Platform Engineering

Production RAG · LLM Systems · Evaluation & Observability | AWS · Python · Spark

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PROFESSIONAL SUMMARY

Senior Data Engineer with 7 years building high-scale data platforms, now building production AI/ML systems with that same platform rigor — evaluation, observability, and grounding. Built a production RAG system (hybrid retrieval, cross-encoder reranking, a 31-question eval harness measuring recall@k and MRR) and LLM tooling connecting AI assistants to a Hive warehouse at a global bank. Seven years making orchestration reliable — 43 pipelines, billions of events/day, latency from days to minutes — now applied to LLM systems.

SKILLS

AI/ML: Production RAG, hybrid retrieval, cross-encoder reranking, embeddings (sentence-transformers), RAG evaluation (recall@k, MRR), agentic workflow patterns, Model Context Protocol (MCP), LLM integration, PyTorch

Data Engineering: Event-driven & metadata-driven architecture, lakehouse (Iceberg), DWH, ETL/ELT, streaming, data modeling, governance & lineage

Cloud / Tools: AWS (Glue, S3, Lambda, EventBridge, Redshift, EMR), Snowflake, dbt, Airflow, Docker/K8s, FastAPI, GitHub Actions, PostgreSQL + pgvector

Big Data: Spark, Hive, Iceberg, Kafka, distributed systems

Languages: Python, SQL, PySpark, Snowpark

PROFESSIONAL EXPERIENCE

Senior Data Engineer — EDP Platform, MUFG Bengaluru | Jan 2024 – Present

AWS · Spark · Python · SQL · Lambda · EventBridge · Iceberg · Glue

- Built an internal Model Context Protocol (MCP) server exposing read-only Hive operations (DDL, partitions, stats, EXPLAIN plans) as tools for AI coding assistants — enabling prompt-based SQL workflows against the enterprise warehouse, with read-only enforcement and data-egress controls.
- Designed serverless ingestion/transform pipelines (AWS DMS, Glue, EventBridge, S3/Iceberg/Redshift, Glue Catalog/Workflows/Triggers) enabling near real-time model risk reporting.
- Built Lambda + boto3 CI/CD deploying 43 Glue workflows (~170 sequential jobs) in ~3 minutes via repeatable, config-driven releases.
- Migrated historical datasets from Redshift to an Iceberg lakehouse for time-travel, auditability, and cost-efficient retrieval.
- Cut pipeline latency from hours to minutes by removing manual hand-offs; standardized validations (row-hash checks, schema-evolution guards).
- Built metadata-driven orchestration with Postgres (RDS) job control + Hive, Redshift, governing stage → transform → enterprise → Info-mart pipelines across multiple risk domains; hardened lineage, migration, and audit/versioning for reproducibility.

Data Engineer — Data Intelligence, SailPoint Technologies Remote (US Team)/Pune | Mar 2022 – Jan 2024

Snowflake · dbt · Airflow · Python · Spark/Snowpark · AWS · CI/CD

- Designed Snowflake-based models powering IdentityAI, processing billions of events/day.
- Built a “Pipeline Factory” to bootstrap new DAGs via config files — reduced onboarding time ~70% and standardized governance patterns.
- Drove cost optimization by replacing Flink with Spark-based Jaccard computations; multi-tenant design processing billions of rows/day, with annual savings in the order of millions.
- Led a Snowpark + dbt POC centralizing testing/data-quality checks; built CI/CD pipelines and migrated internal packages to AWS CodeArtifact for versioned distribution.
- Owned on-call: added alerting/observability, triaged incidents, and coordinated infra fixes for high SLA adherence.

Deputy Manager — Data Science & Engineering, DCB Bank Mumbai | Oct 2019 – Mar 2022

Hadoop (Cloudera/Hortonworks) · Hive · Pentaho · Python · SQL

- Engineered end-to-end pipelines for Internet/Mobile Banking; migrated workloads to Hadoop (Hive) for scalable reporting and lower TCO.
- Automated RBI compliance/operational reporting (financial inclusion, NEFT/RTGS, Video-KYC, ATM dispensations, gold-loan exposure).
- Delivered daily MIS & executive dashboards with validation checks and freshness/quality alerting; mentored 6+ engineers across the Big Data stack.

OPEN-SOURCE PROJECTS

AI Incident Copilot — Production RAG for incident root-cause analysis

- Hybrid retrieval (SQL time-filter + pgvector dense search) with cross-encoder reranking and a confidence quality-gate that suppresses low-grounding answers (no context → no LLM call → no hallucination). FastAPI, Dockerized, CI/CD to Render, 267 tests.
- Built a 31-question evaluation harness (recall@k, MRR) that measured perfect lookup retrieval (recall@5 = 1.0) and surfaced a multi-hop retrieval gap (recall@5 = 0.33) — diagnosed to single-pass retrieval, with scoped fixes (iterative retrieval, chain-aware reranking).
- Chose a deterministic pipeline over an agent for the core flow — incident triage has known steps, and trust under pressure beat autonomous cleverness.

WRITING

From Days to Minutes: Automating Data Pipelines with AWS Event-Driven Architecture — Medium

EDUCATION & CERTIFICATIONS

- AWS Certified Cloud Practitioner
- PG Diploma in Big Data Analytics — CDAC
- Springboard — Data Science (Program)
- B.E. Computer Science — RGPV